

Chapter 1: Living World and Classification of Microbes

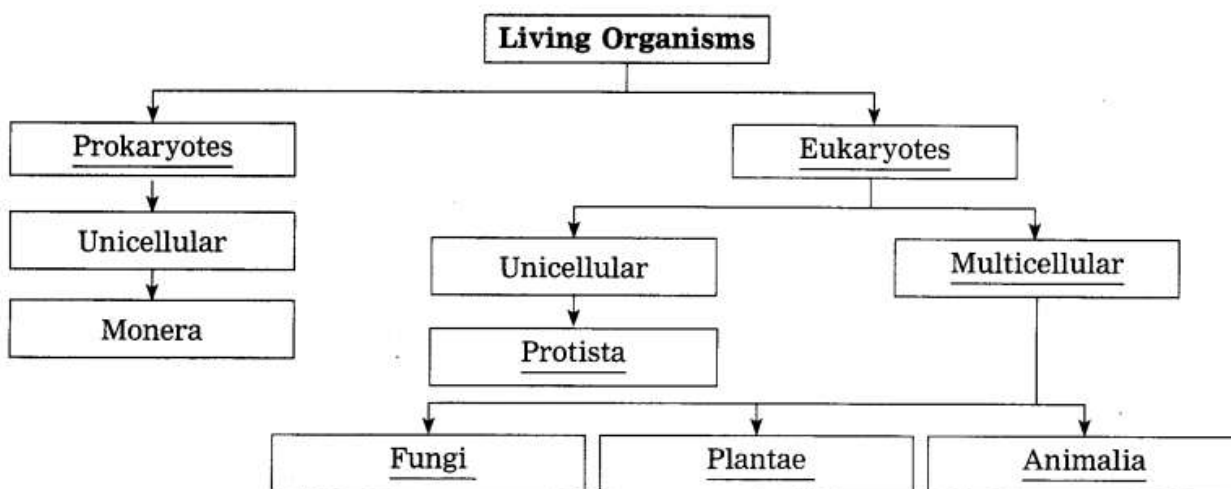
Textbook Exercise Solutions & Answers



1. Use Whittaker method to classify bacteria, protozoa, fungi, algae, prokaryotic, eukaryotic, microbes.

- Bacteria: Kingdom Monera
- Protozoa: Kingdom Protista
- Fungi: Kingdom Fungi
- Algae: If unicellular, Kingdom Protista. If multicellular, Kingdom Plantae
- Prokaryotic: Kingdom Monera
- Eukaryotic: Any kingdom other than Monera.
- Microbes: Kingdom Monera or Protista.

2. Complete the five kingdom method of classification using-living organism, prokaryotes, eukaryotes, multicellular, unicellular, protista, animals, plants, fungi.



3. Find out my partner

A	B
1. Fungi	Candida
2. Protozoa	Amoeba
3. Virus	Bacteriophage
4. Algae	Chlorella
5. Bacteria	Prokaryotic.

4. State whether the following statements are True or False. Explain your statement

(a) Lactobacilli are harmful bacteria.

False.

Explanation Lactobacilli are useful bacteria. They are used for curdling the milk. They are also used for preparing variety of milk products.

(b) Cell wall of fungi is made up of chitin.

True.

Explanation The cell wall of the plants is made up of cellulose while the cell wall of fungi is made up of complex sugar called chitin. Inside the fungal cell wall there is cytoplasm which contains many nuclei.

(c) Organ of locomotion in amoeba is pseudopodia.

True.

Explanation Amoeba is a protozoan, primitive organism which possesses pseudopodia as the organ for locomotion.

(d) Tomato wilt is a viral disease.

True.

Explanation: Tomato wilt is a viral disease. These viruses are able to infect only plant cells.

5. Give answers

(a) State the merits of Whittaker's method of classification.

For the answer of Merits of Whittaker's method refer to point I of answer given on this Digest's page no. 11; under "Use your brain power!"

(b) Write the characteristics of viruses.

1. Viruses are extremely minute of the size range of 10 nm to 100 nm. They are on the borderline of living and non-living. They are 10 to 100 times smaller than bacteria and are visible only through the electron microscope.
2. Though they are not considered as living, they can make replica of their own.
3. They exist in the form of independent particles. They are long molecules of DNA or RNA which are covered by a protein coat.
4. Their survival is possible only inside the living plant or animal cells. When they infect the host cells, they produce their own proteins with the machinery of the host cell. These proteins form many replicas of the infecting virus.
5. This infection destroys the host cell and the virus replicas are released. These free replicas re-infect the other adjoining cells of the host.
6. Most of the viruses are pathogenic, they cause diseases in plants and animals.

(c) Explain the nutrition in fungi.

Most of the fungi are saprophytic in their nutrition. They absorb the nourishment from dead and decaying matter. By taking the organic matter from dead bodies of plants and animals, fungi survive. In this process they degrade the organic matter completely.

(d) Which living organisms are included in the kingdom Monera?

1. All the living organisms belonging to Kingdom Monera are unicellular.
2. The mode of nutrition in them is either autotrophic or heterotrophic.
3. All of them are prokaryotic and thus they do not have well-formed nucleus. The nuclear membrane and the cell organelles are absent in them.
4. Kingdom Monera consists of different types of bacteria and blue-green algae.

E.g. Bacteria *Clostridium tetani*, *Vibrio cholerae*, *Treponema pallidum*, *Streptococcus pneumoniae*, *Legionella pneumoniae*, *Salmonella typhi*, *Staphylococcus aureus*, *Clostridium botulinum*.

6. Who am I?

(a) I don't have true nucleus, cell organelles or plasma membrane.

Microbe from Monera

(b) I have nucleus and membrane bound cell organelles.

Protozoan from Protista/I am any one of Eukaryote

(c) I live on decaying organic matter.

Fungus

(d) I reproduce mainly by cell division. (*Binary fission)

Bacteria and some Protozoa.

(e) I can produce my replica.

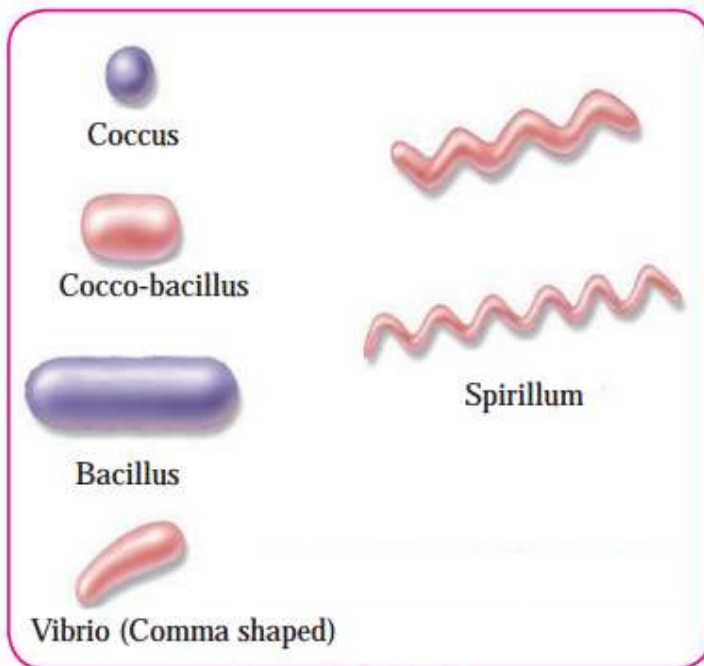
Virus

(f) I am green, but don't have organs.

Algae

7. Draw neat and labelled diagrams.

(a) Different types of bacteria



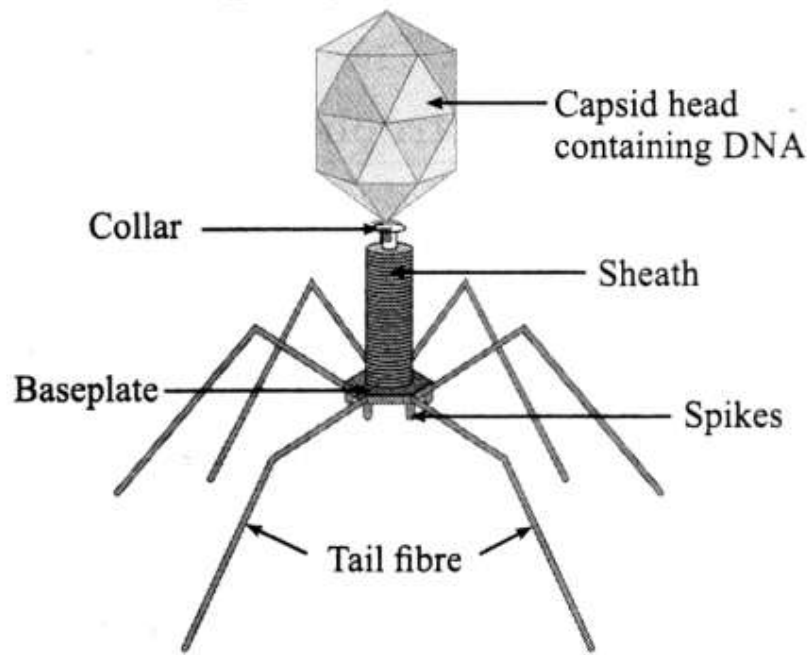
Some Bacteria

(b) Paramecium



Paramecium

(c) Bacteriophage



8. Arrange the following in ascending order of size Bacteria, Fungi, Viruses, Algae.

Viruses, Bacteria, Fungi, Algae.
